

#3D251F
cocap

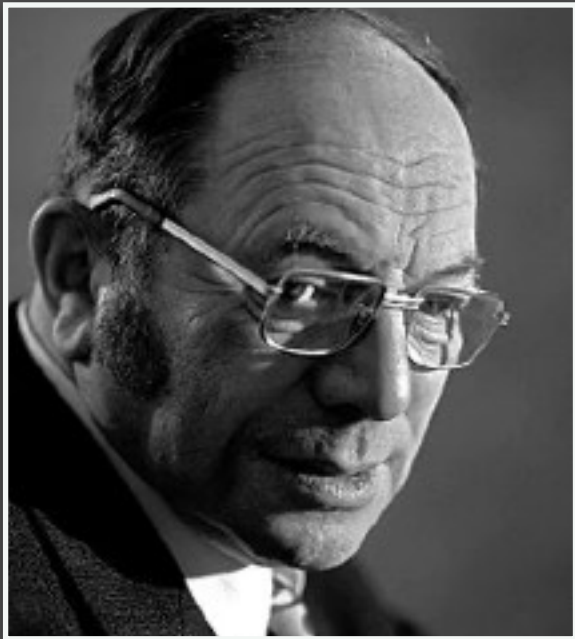
#3DA2B7
teal

#ED6758
salmon

#FEC44C
citrus

#D9B37E
beige

#AA664A
copper



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, d(x, y)$$

Corollary: The optimal dual potentials and φ are related by:

$$\varphi(x) = \frac{1}{2} \|x\|^2 - f(x)$$

$$\psi(x) = \frac{1}{2} \|x\|^2 - g(x)$$

where ψ is cvx and $T^{-1} = \nabla \psi$

Brenier's Theorem

Assumptions:

1. $c(x, y) = \|x - y\|^2$
2. α has a density $\rho_\alpha = \frac{d\alpha}{dx}$



Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

Theorem [Brenier, 1991]:

1. There exists a **unique** coupling π that solves the Kantorovich problem
2. The coupling is deterministic, supported on the graph of a Monge map: $\pi = (\text{Id}, T)_\# \alpha$
3. T is the gradient of a convex function
Formally: the unique $T = \nabla \varphi$ such that φ is **convex** and $(\nabla \varphi)_\# \alpha = \beta$

Corollary: The optimal dual potentials and φ are related by:

$$\varphi(x) = \frac{1}{2} \|x\|^2 - f(x)$$

$$\psi(x) = \frac{1}{2} \|x\|^2 - g(x)$$

where ψ is cvx and $T^{-1} = \nabla \psi$

Applications in ML